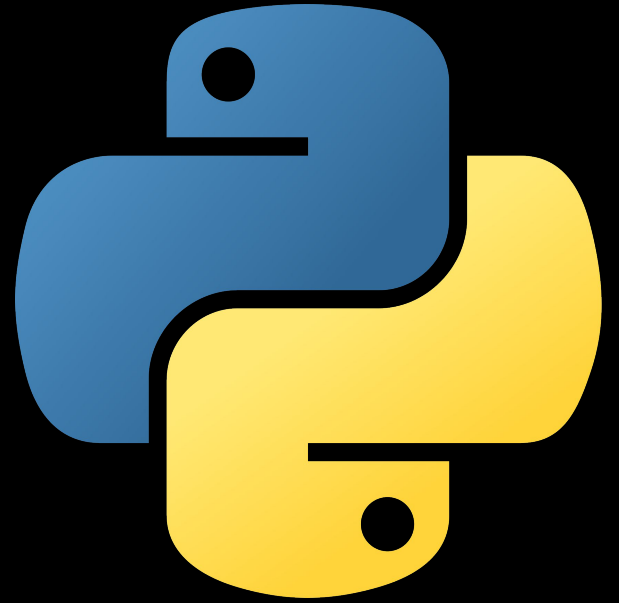


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Pygame Project 2026: Battleships



Session 2

Before we begin...

> If you don't have python installed:

- Go to www.python.org/downloads
- Download and install the appropriate version (not 3.14)
- Follow the instructions below.

> If you have python installed:

- Create a folder with an appropriate name
- Create a new file with an appropriate name

> If you don't have pygame installed:

- Open the terminal
- Type "pip install pygame" (or pip3)

> If you have pygame installed:

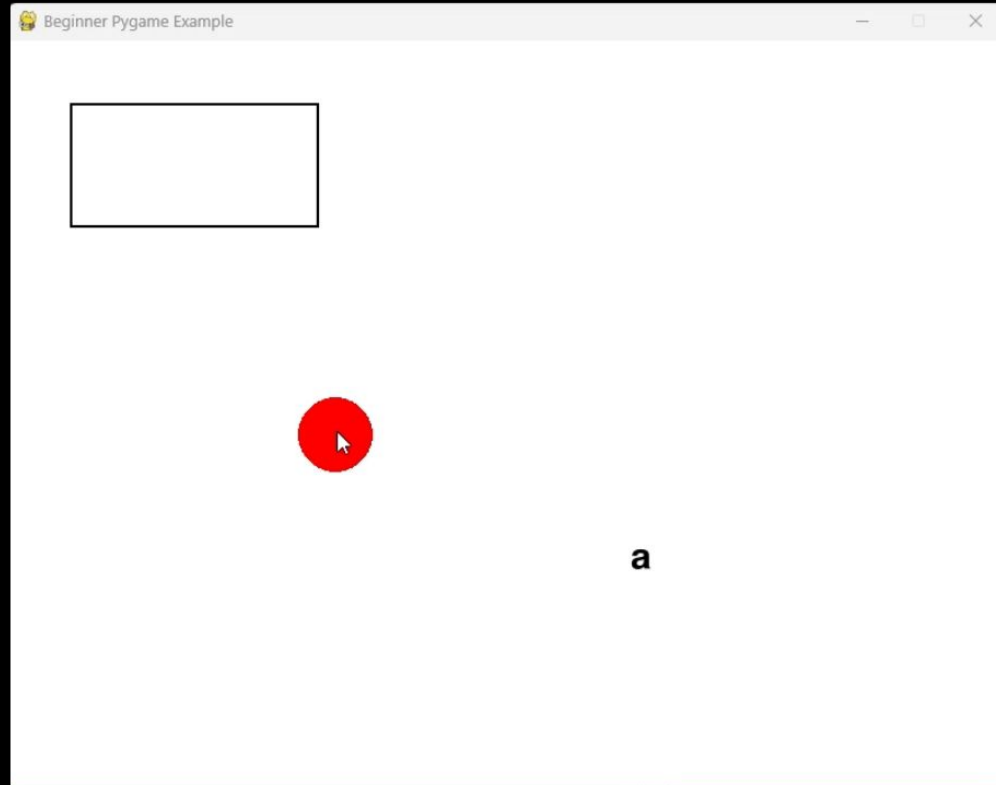
- Make sure it works correctly
- Check correct set up by opening a python shell and typing "import pygame"



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> Today we will use it to create this program:

- > Pygame is a free and open-source cross-platform set of Python modules designed for writing video games.
- > It includes computer graphics and sound libraries designed to be used with the Python programming language.
- > For more information visit <https://pypi.org/project/pygame>



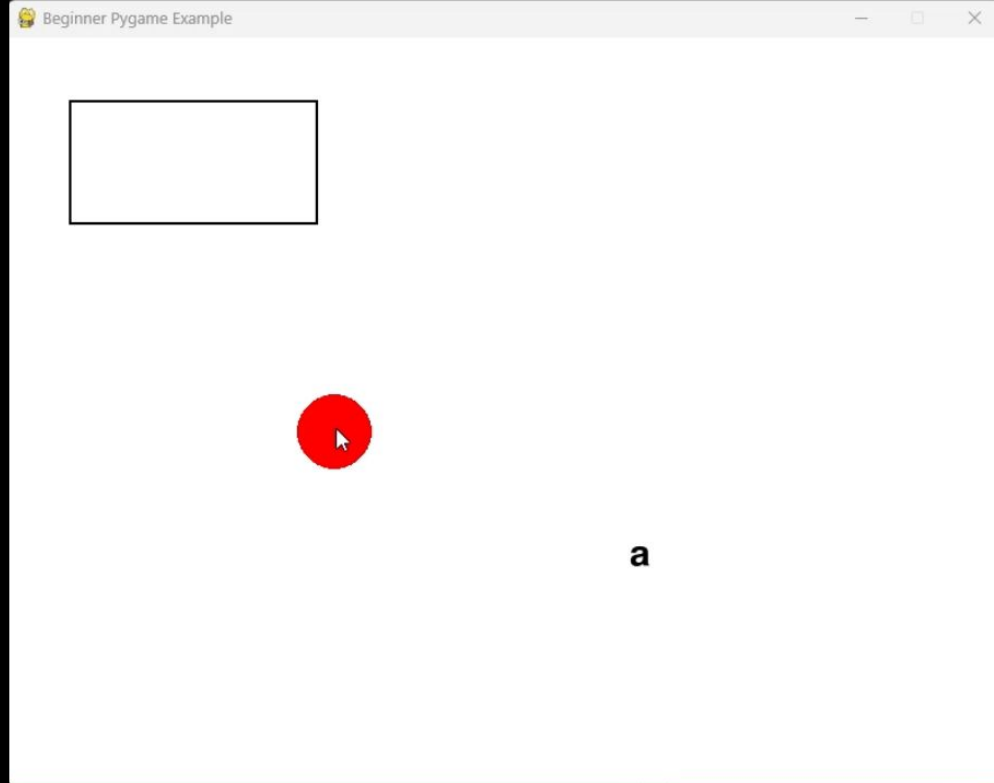


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> Our aim today is to create this program

> It includes:

- Creating a Display
- Drawing Shapes
- Handling Mouse Inputs
- Displaying Text
- Handling Keyboard Inputs



Creating a Display

- > **Pygame uses 'surfaces'**
 - A surface in Pygame is like a blank canvas or a piece of paper where you can draw things, such as images, shapes, or text.
 - Everything you want to display in a Pygame window is drawn on a surface first, and then the surface is displayed on the screen.
- > **There is a surface returned from calling `pygame.display.set_mode(size, flags)` which is the window for your application. Make sure you save this window as a variable.**

> **Remember to import pygame and call `pygame.init()`**



Events

- > Pygame gets inputs as events. For example, any mouse movement, key presses and even the window closing.
- > You can get all current events by iterating through `pygame.event.get()` then checking its type.

- > Remember once `pygame.event.get()` is called, the queue empties.
- > Hints:
 - Try to loop through each event.
 - Use `pygame.K_#` to check for the `#` key.



Drawing Shapes

- > Pygame has a module called draw. This allows the user to draw basic shapes such as:
 - Lines
 - Circles
 - Polygons
 - Rects
- > Pygame's coordinate system is different to a standard mathematical system. (0, 0) is at the top left; as you increase x and y, the point moves to the bottom right corner

> Hints:

- All calls to `pygame.draw` require a surface and a colour to be passed into the function.



Our Basic Loop

> Here are the first few steps for your program:

- Create a display
- Create any Variables
- Start a loop
 - Empty the screen
 - Handle the events
 - Update anything
 - Draw onto the screen
 - Show the screen

> Remember to import pygame and call `pygame.init()` at the start.

> Don't worry. You can ask for help.

> Screen not updating? Make sure you call `pygame.display.flip()` each loop to show your work



Making Text

- > Did you know that you can write text onto the screen?
- > Pygame's module 'font' is used to load a font and create a surface with text onto it.
- > First create a font object with a font name and size. Next call `.render(text)` to get a surface with the text written on it.

> Hints:

- You can get all a list of all available fonts using `pygame.font.get_fonts()`



Making Images

- > Pygame contains a module called `image` that can be used to load images into your program
- > All loaded images are a pixel to pixel to your program.
- > Want to scale or rotate your image? You can use pygame's `pygame.transform` module to do so.

- > To load an image, you need to pass the image's path into the `.load(path)` function.
- > Images taking some time to render? Call `.convert()` on an image to remove unnecessary data. (includes alpha channel)



> **Today:**

- Introduction to pygame
- Creating a display
- Drawing shapes
- Handling inputs
- Writing text

> **Next Session:**

- Applying pygame into Battleships
- Playing our game

Thank you for coming!

See you next week.